FEASIBILITY ASSESSMENT

Feasibility Report

Technical, operational and delivery feasibility of the proposed prototype

FEAS-02

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VAYU NEXA AI | Alchemy Architects | SIH 2026

Prepared 20 October 2026 | Design assessment, not a certification

1. Executive verdict

AAKAASH ADHINAYAK is feasible as a bounded student software prototype if its scope is sequenced around a functioning generic scheduling engine, import/validation, independent constraint verification, reproducible scenario comparison and one real public-data integration. A production-grade multi-source decision platform with predictive maintenance, persistent security, grounded AI and polished 3D analytics is a much larger effort. Feasibility should therefore be judged by the minimum credible demonstrator, not by the maximum feature wish-list.

Evidence note: The public URL resolves to a page titled "AAKAASH ADHINAYAK - Verified Intelligence for Adaptive Resource Planning", but the available research interface did not expose navigable page text, runtime interactions, source code, API traces or test results. Therefore this report distinguishes observed deployment metadata and the team-supplied blueprint from proposed design and unverified implementation. It is not a source-code audit, security assessment, or certification of the live system.

2. Feasibility matrix

Dimension	Assessment	Reason	Core mitigation
Web architecture	High	React/TypeScript with typed APIs is conventional; AI Studio supports full-stack deployments on Cloud Run [R2].	Confirm exportable code, available server runtime and deploy limits.
Constraint optimisation	Medium-High	CP-SAT supports constraint modelling but implementation requires careful integer modelling, status mapping and tests [R6].	Start with a generic assignment/scheduling model and independent validator.
Public weather integration	Medium	API is documented, but rate limits and CORS restrictions call for a server proxy [R3].	One endpoint first; timeout, retry/backoff, cache and stale-state tests.
Predictive maintenance	Medium	NASA PCoE catalog contains candidate datasets, but each exact archive must be obtained and reviewed [R5].	Dataset feasibility gate before training; report baseline metrics only after test.
Grounded LLM assistant	Medium	Tool-mediated retrieval is feasible; safe grounding, auth and hallucination tests are necessary [R7].	Initially read-only; each answer must cite actual record IDs.
Interactive 3D visualisation	Medium	Three.js can depict record/run relationships, but high-cardinality visuals and mobile GPU limits create performance risk.	Graph only a selected subgraph; include a 2D/accessible fallback.
Team execution	Medium-Low	Project history indicates one principal application developer, with other members contributing research, data and design.	Time-box tasks; produce a narrow end-to-end vertical slice early.
Operational defence data	Not feasible without authority	The reviewed public sources do not establish access to live defence assets, personnel or weapon-system data.	Keep operational integration out of scope; use generic/synthetic planning fixtures.

3. Scope that is feasible

Minimum credible demonstration

- Upload a documented CSV or JSON file.
- Show validation findings for missing, malformed, duplicate, stale and conflicting records.
- Create a generic scenario and run a deterministic solver.
- Independently validate the candidate solution, including infeasible outcomes.
- Trigger a controlled synthetic disruption and compare pre/post runs.
- Export an evidence report that includes inputs, source labels, solver state and validation results.
- Show one actual public-data connector with timestamps and correct error handling.

Stretch scope after the core passes

- Predictive-maintenance model experiment from an obtained public dataset.
- Evidence-grounded VAYU NEXA AI with typed read-only tools.
- Three.js graph linked to the same source records and run IDs.
- Persistent accounts, reviewer workflows, audit history and role-level permissions.
- Automated load/performance and accessibility checks.

4. Effort estimate and team operating model

The estimates below are planning allowances, not observed delivery times. They assume a responsive web app, a single principal developer, team members able to own research, data preparation, visual design and testing, and limited external integration. Calendar time depends on availability, code quality, and the condition of the existing deployment.

Workstream	Indicative effort	Owner emphasis	Dependency
Audit current app and recovery/export	1-3 focused days	Aasidh + technical helper	Access to project source and deployment.
Schema, import and data-quality suite	4-7 developer-days	Aasidh; Harini/Saran define data tests	Stable schema and sample fixtures.
Solver + independent validator	5-10 developer-days	Aasidh; Saran supports model/data analysis	Constraint specification and test cases.
Scenario history and evidence export	3-6 developer-days	Aasidh; Majitha reviews report clarity	Stable run and validation schemas.
Public API connector	2-5 developer-days	Aasidh; team verifies sources	Endpoint access and server environment.
Predictive experiment	4-8 developer-days	Aasidh + dataset support from team	Dataset availability, terms and baseline.
AI + 3D integration	5-12 developer-days	Aasidh + Ajithram visual design	Stable APIs; bounded visible graph.
Security, accessibility, demo hardening	4-7 developer-days	Whole team with checklists	Feature freeze and test environment.

The streams overlap and should not be summed mechanically into calendar time. The recommended strategy is a 2-3 week proof-of-engineering sprint for the minimal core, then short additional sprints for data science, AI, 3D and hardening. If SIH submission or demo timing is tighter, drop stretch scope rather than falsify implementation status.

5. Data and deployment feasibility

AviationWeather.gov supports machine access to aviation weather, yet documents limited historical windows, a 100-request-per-minute limit, maximum result counts on many endpoints and no browser CORS support [R3]. Build a server-side connector with request scoping, user-agent identification, short TTL caching, rate-aware retries, source timestamps and graceful unavailability. Do not expose secrets in the client.

Google AI Studio custom URL deployment uses a Cloud Run service; the official deployment guide describes tier and billing prerequisites [R2]. Cloud Run cost depends on region and usage and the free tier is conditional [R12]. Budgeting must include application compute, database reads/writes, logs, network transfer, model/API usage, storage and potential paid tiers. Do not assume a custom URL means unlimited free production service.

6. Principal risks and control plan

Risk	Likelihood / Impact	Mitigation
Current source code cannot be exported or reconstructed	Medium / High	Secure export or repository access before major edits; retain a release snapshot.
Solver and UI disagree	Medium / High	Independent validator, shared typed contracts, run-specific snapshots and regression tests.
Live API unavailable or stale	Medium / Medium	Cache, explicit timestamps, error states and a labelled offline demonstration.
Dataset inaccessible or licence uncertain	Medium / High	Inspect exact archive and terms before selecting a model; keep a fallback dataset.
LLM fabricates facts or overstates results	Medium / High	Retrieval-only evidence tools, citations/IDs, “unknown” behaviour and adversarial tests.
3D interface slows core tasks	Medium / Medium	Bounded graph, lazy loading, reduce motion, 2D mode and performance budgets.
Feature inflation prevents completion	High / High	Set a feature freeze after core acceptance tests pass.

7. Feasibility decision

Decision: PROCEED WITH CONDITIONS. The core student prototype is feasible. The complete target vision is only conditionally feasible and must be staged. The first go/no-go gate is access to the source code and proof that import, optimisation, independent validation, replanning and evidence export work together. The public deployment alone is not enough evidence to pass that gate.

References: [R2], [R3], [R5], [R6], [R7], [R10], [R12]. See the separate References Report for full source URLs.